

Operator Algebraic Structures I

The Process of Quantization

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1 Introduction

We'll introduce the contents of this text and what you should know before reading it through a list of questions which will be answered. These questions are some of the “usual” ones that any reader would have before reading any text.

1.1 What will you cover?

1.2 What are the prerequisites?

1.3 Where can I go after working through this text?

1.4 What is the historical backdrop of this subject?

1.5 Conventions & General Structure

Most of the structure of the thesis is clear/self-explanatory, so we won't bother getting into the details. There is one major way in which we've re-organized the contents of the thesis that is different from other books/theses and we do want to explain it. Throughout the thesis, you will see boxes of the following form:

1. Definition boxes.
2. Lemma boxes.
3. Proposition boxes.
4. Theorem boxes.
5. Remark boxes.
6. Intuition boxes.
7. Notation boxes.
8. Terminology boxes.
9. Claim boxes.
10. Proof boxes.
11. Motivation boxes.
12. Miscellaneous Results boxes.

Most of these are self-explanatory and we use them whenever we want to note down something that might get potentially referenced again later. This allows us to use the internal referencing feature of LaTeX. The last one requires further explanation. When I was writing this thesis, I intended to go through parts of the literature and fill in details as much as possible. On the other hand, presenting **all** of those details in the main body of the thesis would've been impractical. Moreover, it creates unnecessary difficulties for the reader, particularly those who rely on authors to emphasize what is important and be "quick" about that which isn't so crucial. So, we've successfully found a middle-ground; by placing these "small" (but sometimes, quite technical and difficult) proofs within these miscellaneous results, we create a setting where they can be referenced again if necessary but do not break the narrative structure of the thesis. This should make the thesis much easier to read; one can simply understand the miscellaneous results and move on to the main proofs, perhaps even trying to prove the miscellaneous results themselves.

The best part about the way we've handled this is that these miscellaneous results boxes are directly linked to the corresponding proofs which are located towards the end of the thesis. One simply needs to click the title of a given box (say, **Misc. Res. 4.2.3**) and one is immediately taken to the corresponding miscellaneous proof for that result. Conversely, one can go back to the result from the proof by clicking the title of that proof box. Note that a similar feature does not work for all of the other boxes. This is why we highly recommend having a digital copy of the thesis at hand even if one wishes to use a paper version.

Now, we also have a word or two about standard notation that will be used throughout the thesis. For the most part, there should be no major contradictions between the notation we present here and that which is typically used throughout the thesis. For the sake of clarity, we'll split these up into several boxes that deal with different aspects of the conventions and notation that we'll use. We've chosen these conventions so that they're pretty easy to understand without having to explicitly refer to our choices here. Nevertheless, we have them here for easy reference in the future.

General Mathematical Notation

$\mathbb{K}$ denotes either $\mathbb{R}$ or $\mathbb{C}$.

$\mathbb{F}$ denotes any field.

$\mathbb{N}$ denotes the set of natural numbers (excluding 0)

$\mathbb{N}_0$ is the set of natural numbers including 0

$\mathbb{R}_0^+$ denotes the set of all non-negative real numbers.

$\mathbb{R}^+$ denotes the set of all positive real numbers.

If X, Y are sets, then $\text{Maps}(X, Y)$ is the set of all functions from X to Y .

If A, B are mathematical objects, then $A \stackrel{\text{def}}{=} B$ means that A is defined to be equal to B .

If A, B are statements, then $A \stackrel{\text{def}}{\Leftrightarrow} B$ means that A is defined to be equivalent to B .

If $G \in \text{Grp}$ and H is a subgroup of G , then we'll denote it by $H \leq G$.

If $G \in \text{Grp}$ and H is a proper subgroup of G , then we'll denote it by $H < G$.

If $G \in \text{Grp}$, then:

$$Z(G) \stackrel{\text{def}}{=} \{z \in G \mid \forall g \in G : zg = gz\},$$

is the center of G .

Here are some common acronyms we'll use.

Common Acronyms

WLOG - Without loss of generality.

TFAE - The following are equivalent.

TFSH - The following statements hold.

nbhd - neighborhood.

Here are some further general notes for the reader before one begins to read this text.

Notes For The Reader

The reader of these notes should keep the following statements in mind.

1. Any and all mistakes in these notes are my responsibility. If you spot any of these, please contact me immediately at a.vats.math (at) gmail.com^a and let me know so I can fix those mistakes.
2. If you see question marks at any point, then one of the following issues might have occurred:
 - (a) There's something being cross referenced in the text but I haven't used the correct label.
 - (b) There's something in another document which is being cross referenced.
 - (c) There's something that's being cross referenced which doesn't exist.

Whatever the case, do let me know about them because they need to be fixed.

3. If you find a result in this document which does not have a complete proof or anything like that, then it means that I probably just got lazy while writing the proof and haven't gotten back to it yet. I will eventually return to these and fill in the details but it's great if you point them out explicitly.
4. If you find a result in this document for which I've only provided a reference rather than a proof, it's likely because I got quite lazy and postponed writing that proof for a later time. In that case, feel free to inform me about it so I'm forced to actually work through the proof. Moreover, if you have suggestions for improvements to proofs or more conceptual proofs, feel free to reach out as well, you will be appropriately credited.
5. The document makes heavy use of the Cref feature. Of course, you can print whichever bits of it as you want but it's good to have an electronic copy pulled up as well so you may optimize your experience using it.
6. There are certain pdf readers which allow you to hover over a hyperlink in the pdf and get a preview of what's in that hyperlink. This allows you to simply glance at what's in there (which is all that one needs to do sometimes) without having to click the link & actually go to the source. I've listed a couple of pdf readers:
 - (a) For Linux, use GNOME document viewer/Evince.
 - (b) For MacOS, use Skim.
 - (c) For Windows, use Sioyek.

I strongly recommend using these as they vastly improve the experience of reading these notes digitally.

7. The credit for the latex template goes to Senan Sekhon.
8. The credit for the cover page goes to <https://ar.pinterest.com/pin/331999803788929234/>.

If you would like to:

1. contribute to the material in here as an author;
2. comment on the material in here;

then please contact Abhijeet Vats so that further possibilities for collaboration may be discussed.

^aYou may refer to him "Abhi" or "Mousedorff" or any other cool nickname of your choosing.

2 Several Sales Pitches

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

Let's start with some motivation.

| **Motivation.**

3 The Structure of Banach Algebras

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some "motivation".

| **Motivation.**

4 Spectra & Invertibility

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

Let's start with some motivation.

| **Motivation.**

5 Gelfand's Theory of Commutative Banach Algebras

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

6 Some Important Computations of Gelfand Spectra

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

7 Topological (Unital) Involutive Structures

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

8 The Rigid Structure of C^* -Algebras

Summary

The following topics will be discussed in this lecture.

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

We begin with some motivation.

| **Motivation.**

9 Gelfand-Naimark Duality

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

Let's start with an overview.

| **Motivation.**

10 Positivity on C^* -algebras

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

11 Homological Aspects of C^* -Algebra Theory

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

12 States, Traces & Some Simple C^* -Algebras

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

The motivation for this lecture will be given throughout it, so we will not provide separate motivation here.

13 Towards C^* -algebraic Representation Theory

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

Let's start with some motivation.

| **Motivation.**

14 Spectral Theory I: Functional Models

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

There are no special conventions for this lecture.

Let's start with some motivation.

| **Motivation.**

15 Constructing Matrix C^* -Algebras

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

If $V, W \in \text{Vec}_{\mathbb{C}}$, then their algebraic tensor product is denoted by $V \otimes_{\mathbb{C}} W$.

Let's start with some motivation.

| **Motivation.**

16 Who Gives AF? Inductive Limits of C^* -Algebras

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

17 Discrete Group C^* -Algebras

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

18 Introducing C^* -Module Theory

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

19 Introducing Four Operator Topologies

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

20 Convexity: The Key Property

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

21 Applications of Convexity Theory

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

22 The Bicommutant Theorem & Crucial Examples

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

23 The Structure of Abelian vN Algebras

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

24 Spectral Theory II: Integral Decomposition

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

25 The Kaplansky Density Theorem & Duality

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

We've provided enough motivation throughout the last lecture. You should read the beginning of ?? for further directions on how to read this lecture, as there is a bunch of material here that cannot be internalized all at once.

26 Preduals & Sakai's Theorem

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

27 Aspects of Noncommutative Measure Theory

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

28 Three Crucial Properties of Analytic Group Theory

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

29 A Weaker Form of Positivity: Towards Connes' Theorem

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

We will provide some lengthy motivation in ??, so we will not provide separate motivation here.

30 Wrapping Up & Introducing Loose Ends

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

31 Soficity, Hyperlinearity & Graph Approximations

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

32 The Connes Embedding Problem: Building Up Towards Kirchberg's Theorem

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

33 Towards The Classification Theory of Type III Factors

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation and, in fact, a good chunk of this lecture will simply be about motivation.

| **Motivation.**

34 Spectral Theory III: Multiplying The Magickery

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

35 Spectral Theory IV: Closed Forms & Integral Decomposition

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

36 The Properties of Modular Automorphism Groups

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

37 Constructing Tensor Products

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

38 Nuclearity & Exactness I: Some Barebones Basics

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

If $V, W \in \text{Vec}_{\mathbb{C}}$, then their algebraic tensor product is denoted by $V \otimes_{\mathbb{C}} W$.

Let's start with some motivation.

| **Motivation.**

39 Nuclearity & Exactness II: Deeper Ideas & Facts

Summary

Here are the (less common) notational conventions we'll follow for this lecture.

Conventions

Let's start with some motivation.

| **Motivation.**

Solutions

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